AI-BASED PHISHING DETECTION FRAMEWORK

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WHY PHISHING DETECTION MATTERS ?

The Growing Cybersecurity Challenge

- Millions of phishing websites are created every year.
- Fake websites imitate trusted platforms to deceive users.
- Financial institutions, e-commerce sites, and social media accounts are common targets.
- Existing blacklist methods cannot always detect newly created phishing websites.

PROPOSED SOLUTION

AI-Based Detection Framework

Instead of relying on manually maintained blacklists, the system analyzes the characteristics of a URL and predicts whether it is safe or malicious using Machine Learning.

MAIN COMPONENTS

- User-friendly web interface
- Feature extraction engine
- Machine Learning prediction model
- Risk assessment module
- Result display

PROJECT WORKFLOW

How the System Works

1. User enters a website URL.
2. URL features are extracted automatically.
3. Features are processed by the trained AI model.
4. The model predicts whether the website is legitimate or phishing.
5. The result is displayed with the corresponding risk level.

BUILDING THE AI MODEL

machine Learning Pipeline

- Data collection from phishing and legitimate URL datasets
- Data cleaning and preprocessing
- Feature engineering
- Model training
- Performance evaluation
- Model deployment

why Random Forest ?

- High classification accuracy
- Good handling of multiple features
- Reliable performance on cybersecurity datasets

SYSTEM DEVELOPMENT

Technology Stack

Layer	Technology
Frontend	React.js
Backend	Flask
Machine Learning	Scikit-learn

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Data Processing	NumPy
API	REST API
Deployment	Vercel & Render
Version Control	GitHub

IMPLEMENTATION HIGHLIGHTS

Features Developed

- Secure URL scanning page
- Dashboard for quick access
- Instant prediction results
- Responsive user interface
- Backend API integration
- Cloud-hosted application

TESTING & PERFORMANCE

System Evaluation

The application was tested for :

- Prediction accuracy
- API functionality
- Response time
- User interface responsiveness
- Frontend-backend integration

Outcome

- Real-time predictions
- Reliable classification
- Stable application performance

KEY LEARNINGS & FUTURE ENHANCEMENTS

Skills Gained

- Machine Learning model development
- React application development
- Flask API integration
- Cloud deployment
- Git and GitHub collaboration

Future Enhancements

- Browser extension
- Email phishing detection
- QR code analysis
- Deep Learning models
- Live threat intelligence integration

CONCLUSION

- Demonstrated how Artificial Intelligence can strengthen phishing detection.
- Successfully integrated Machine Learning with a full-stack web application.
- Developed a scalable framework capable of supporting future cybersecurity enhancements.

THANK YOU